Measuring the Local Effects of Typhoons on Economic Activities

Proposed chapter for the GIE textbook

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Motivation

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- Traditional economic data fail to capture localized, short-term impacts.
- Nighttime lights and cyclone track data allow high-frequency, spatially granular assessment.
- Vietnam offers a case study with rich storm exposure and rapid economic change.

Chapter objective

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- Present practical guidance for combining nightlight data with storm tracks.
- Compare datasets and econometric strategies.
- Discuss generalizability, spatial spillovers, and methodological caveats.

Data at a Glance

Nighttime Light:

- DMSP-OLS (1992–2013): Coarse, top-coded.
- VIIRS-DNB (2012–present): High-res, calibrated.

Typhoon Exposure:

- IBTrACS + JMA: Tracks, wind speed, timing.

Vietnam: A Case Study

- 79 typhoons (1992–2020) with varying intensity.

Vietnam: A Case Study

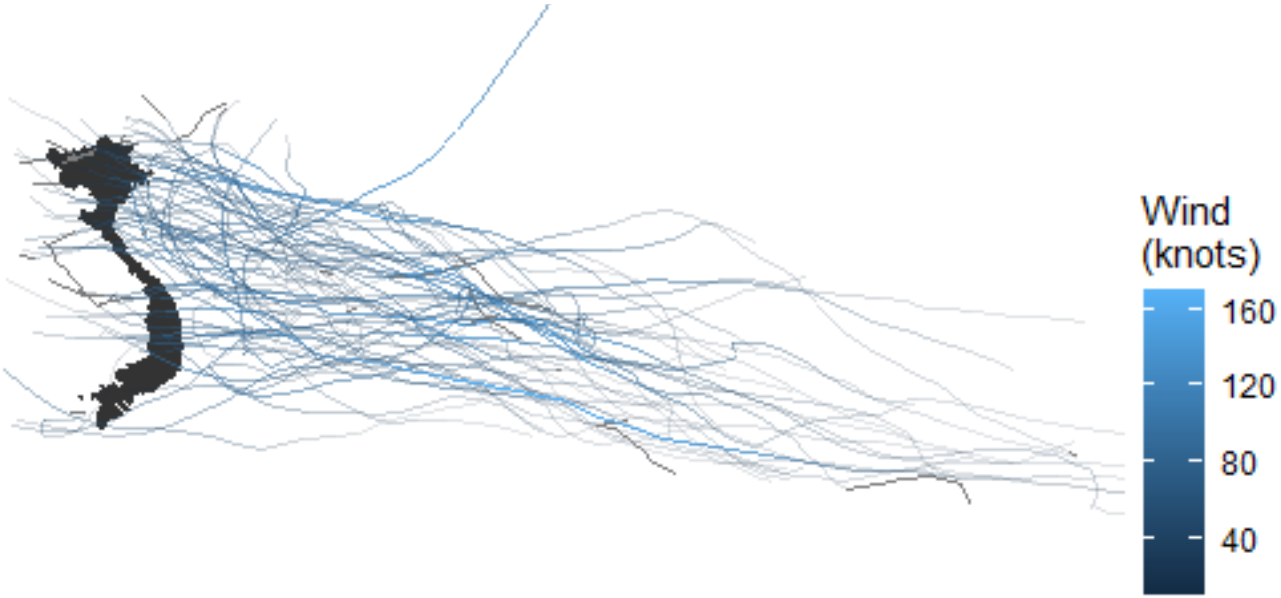
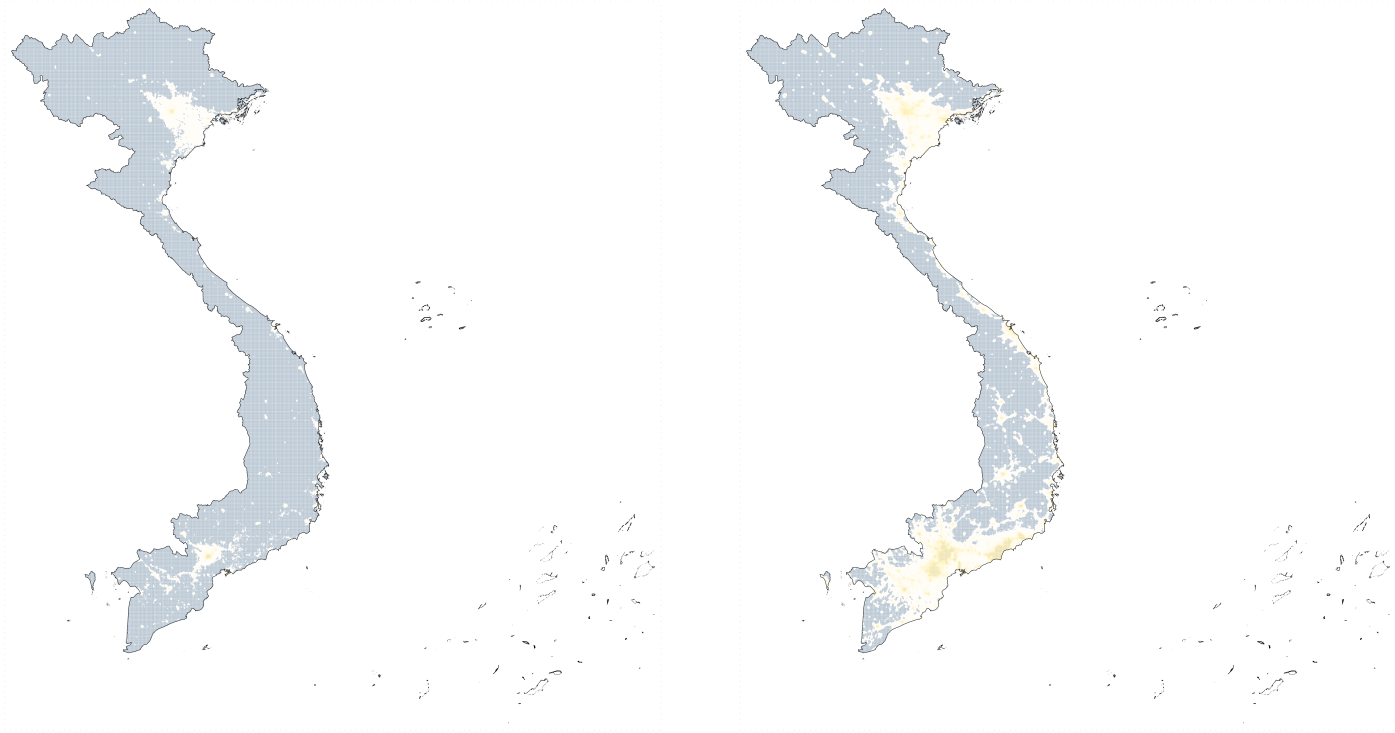


Figure 1: Typhoon tracks between 1992 and 2020

Vietnam: A Case Study

- 79 typhoons (1992–2020) with varying intensity.
- Combined with 1km grid panel of nighttime lights.

Vietnam: A Case Study



(a) Nighttime light in 1993 **(b)** Nighttime light in 2020

Figure 2: Nighttime light intensity across Vietnam (1993 vs 2020)

Vietnam: A Case Study

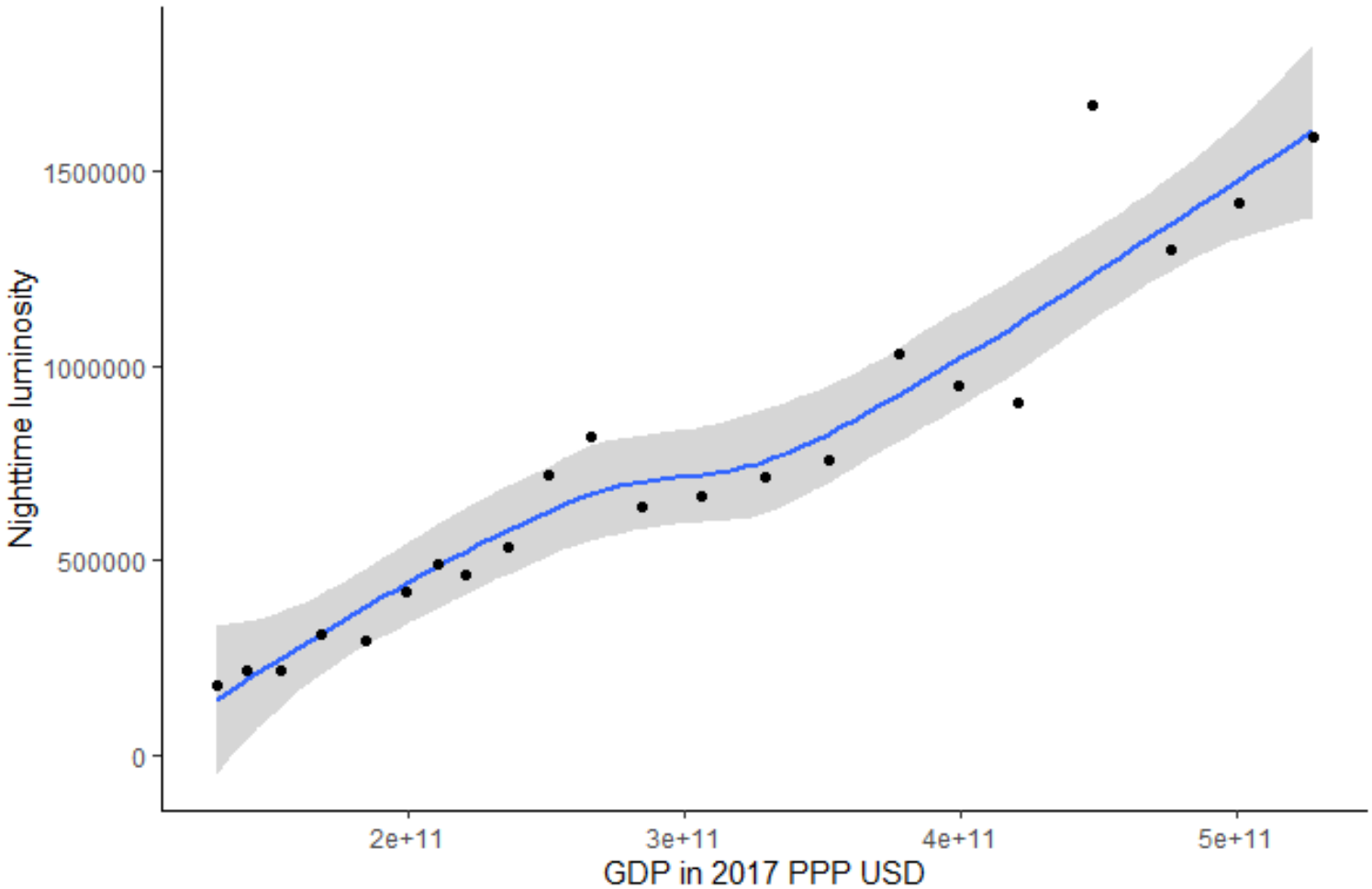


Figure 3: Correlation between GDP in 2017 USD and Nighttime luminosity

Vietnam: A Case Study

- 79 typhoons (1992–2020) with varying intensity.
- Combined with 1km grid panel of nighttime lights.
- Track frequency and duration of exposure across grid.

Vietnam: A Case Study

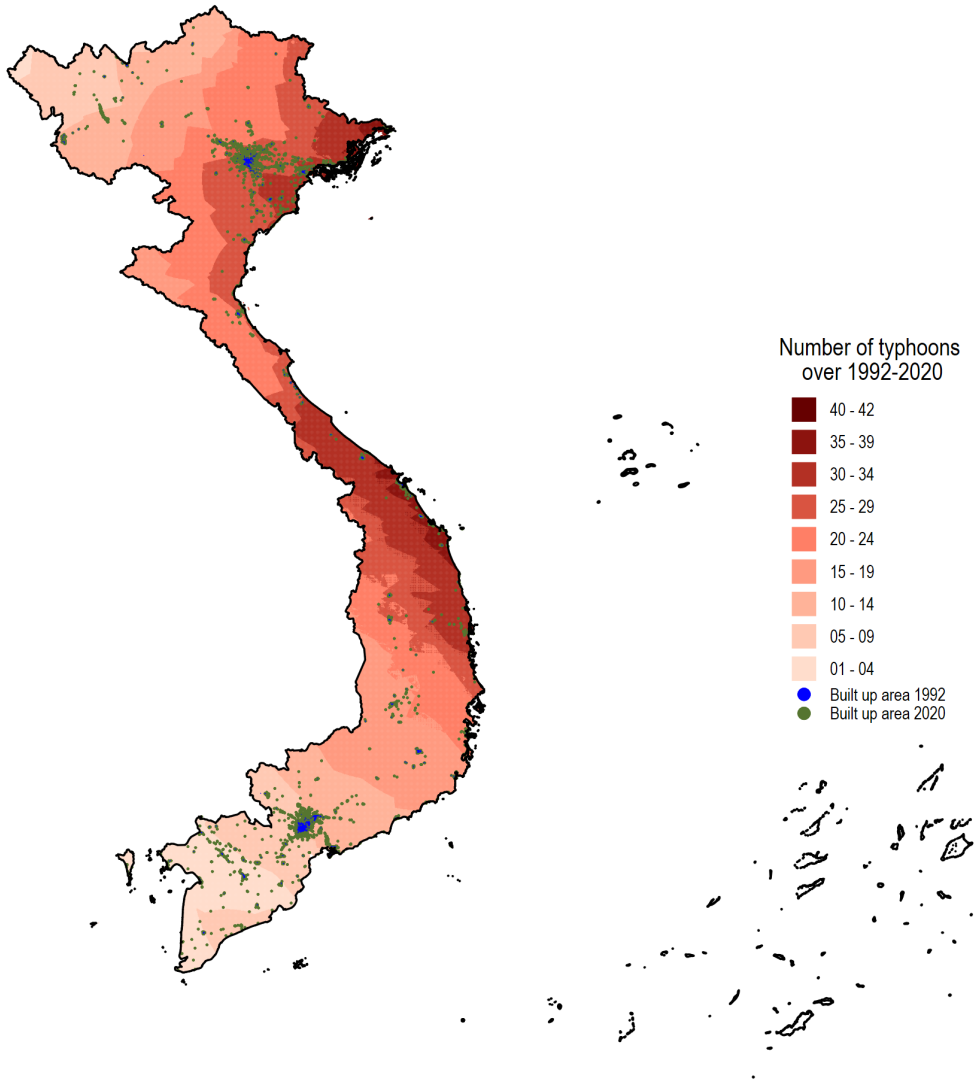
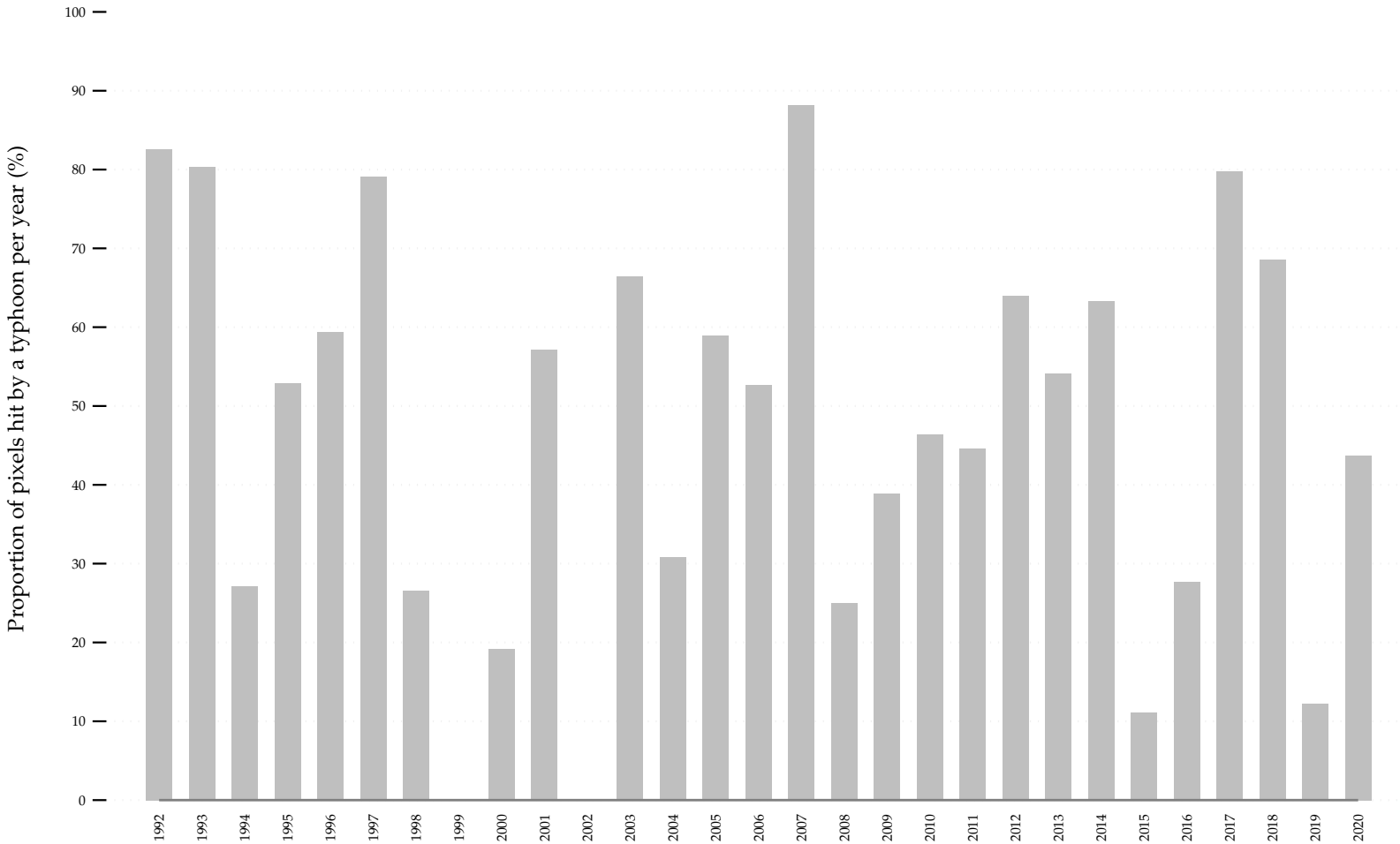


Figure 4: Number of typhoon hits (1992–2020)

Vietnam: A Case Study



Source: Authors calculation based on IbTracks datasets.

Figure 5: Proportion of locations hit by typhoons (1992–2020)

Econometric Framework

- Track changes in nighttime lights post-typhoon using DiD.
- Dynamic models (event study) to capture persistence.
- Estimators from Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024).

Preview based on DMSP-OLS

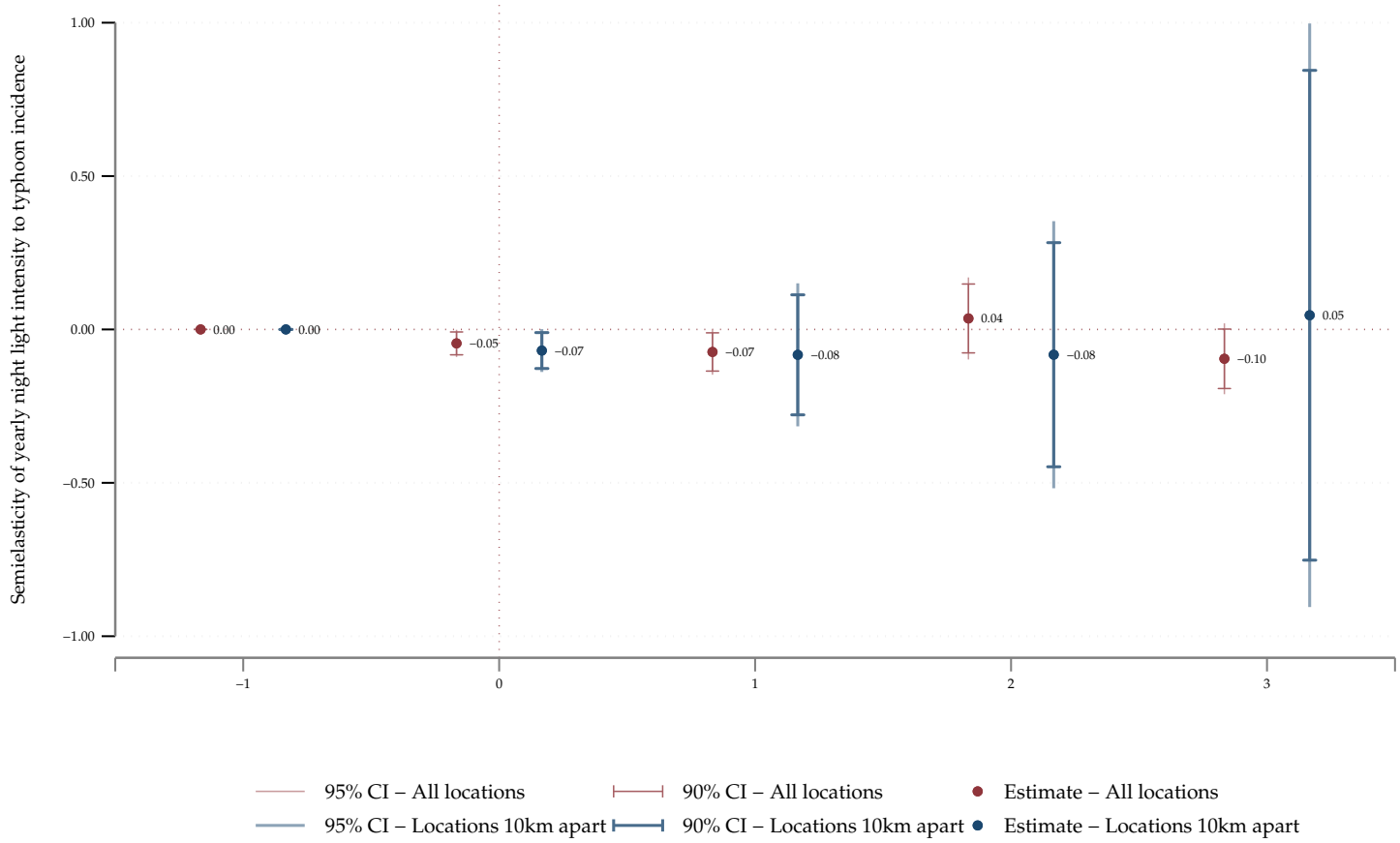


Figure 6: Nightlights drop by 5–8% post-typhoon

- Effects persist up to 2 years post-landfall.

Data Comparison & Spatial Considerations

Nightlight Datasets:

- **DMSP:** Longer series but more noise.
- **VIIRS:** Shorter but cleaner and more granular.
- Same model run on both to compare sensitivity.

Spillovers and Spatial Effects:

- Typhoon shocks may spill across locations.
- Consider indirect effects on neighboring areas not directly hit.

Why This Chapter Matters

- Guides GIE users through practical use of global open data.
- Elevates disaster impact evaluation beyond household surveys.
- Encourages replication and extension to new settings.

To go further

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Related works:

1. Espagne E., B. Y. Ha, K. Houngbedji and T. Ngo-Duc, 2022. "Typhoons and urbanisation in Vietnam", *Revue d'économie du développement* 31: 57-62.
2. Espagne E., B. Y. Ha, K. Houngbedji and T. Ngo-Duc, 2022. "Effect of Typhoons on economic activities in Vietnam: Evidence using satellite imagery", AFD Research Papers 263

References

Callaway, B. and P. H. Sant'Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics 225*(2), 200–230.

de Chaisemartin, C. and X. D'Haultfœuille (2024, 02). Difference-in-Differences Estimators of Intertemporal Treatment Effects. *The Review of Economics and Statistics*, 1–45.